

Task 1

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, Dense, MaxPooling2D, Flatten
```

Task 2

```
model = Sequential([
    Conv2D(32, kernel_size=3, activation='relu', input_shape=(28, 28, 3)),
    MaxPooling2D(),
    Flatten(),
    Dense(64, activation='relu'),
    Dense(1, activation='softmax')
])
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
model.summary()
```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input\_shape`
super().__init__(activity_regularizer=activity_regularizer, **kwargs)

Model: "sequential_3"

Layer (type)	Output Shape	Param #
conv2d_15 (Conv2D)	(None, 26, 26, 32)	896
max_pooling2d_6 (MaxPooling2D)	(None, 13, 13, 32)	0
flatten_2 (Flatten)	(None, 5408)	0
dense_10 (Dense)	(None, 64)	346,176
dense_11 (Dense)	(None, 1)	65

Total params: 347,137 (1.32 MB)

Trainable params: 347,137 (1.32 MB)

Non-trainable params: 0 (0.00 B)

```
print(f"No of Convolutional Layers : 1")
print(f"No of Dense Layers : 2")
print(f"No of Pooling Layers : 1")
```

```
No of Convolutional Layers : 1
No of Dense Layers : 2
No of Pooling Layers : 1
```